

ECE 4375, Midterm Exam (March 5th, 2025)

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(87)

Do not open the exam until you are told to start

Notes:

1. For full credit, the final answer **must** be clearly marked and work shown
2. Only a calculator, writing instruments, and 1 crib-sheet is allowed
3. Do not share calculators, pens, paper, notes, or erasers during the exam
4. Do not unstaple the exam, use only paper provided
5. Place additional worksheets (numbered and sorted) in the exam
6. Do not continue writing when time is called.
7. **Round all calculations and solutions to the hundredths place**

EQ1: $Y = E_2 H_2 H_3 - E_1 G_1$

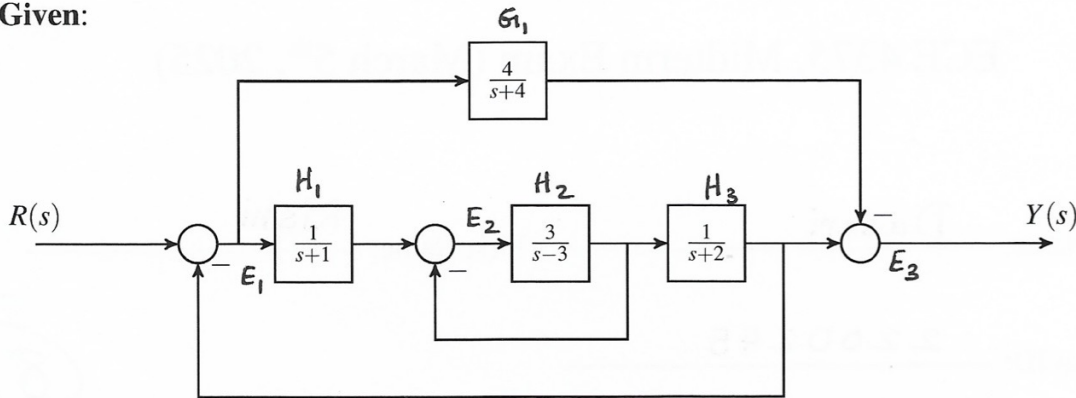
EQ2: $E_2 = E_1 H_1 - E_2 H_2$

EQ3: $E_1 = R - E_2 H_2 H_3$

T-1:15

T-0:32

1. Given:



Find:

A. the transfer function, $\frac{Y(s)}{R(s)} = \frac{B(s)}{A(s)}$

B. from part A determine the loop gain, L

A.) $Y = H_3 H_2 E_2 - G_1 E_1$: EQ1

$E_2 = H_1 E_1 - H_2 E_2$

: EQ2 $\Rightarrow E_2 (1 + H_2) = E_1 H_1 \Rightarrow E_1 = \frac{E_2 (1 + H_2)}{H_1}$

$E_1 = R - H_3 H_2 E_2$

: EQ3

$\Rightarrow E_2 = \frac{E_1 H_1}{1 + H_2}$

~~$\Rightarrow Y = H_3 H_2 \left(\frac{E_1 H_1}{1 + H_2} \right) - G_1 E_1$~~

$\Rightarrow \frac{E_2 (1 + H_2)}{H_1} = R - H_3 H_2 E_2$

$\Rightarrow R = E_2 \left(H_2 H_3 + \frac{(1 + H_2)}{H_1} \right)$

$\Rightarrow R H_1 = E_2 (H_1 H_2 H_3 + 1 + H_2)$

$\Rightarrow E_2 = \frac{R H_1}{(H_1 H_2 H_3 + H_2 + 1)}$

$\Rightarrow E_1 = R \left(1 - \frac{H_1 H_2 H_3}{1 + H_2 + H_1 H_2 H_3} \right)$

T-1:05
T-0:29 $Y = \frac{H_1 H_2 H_3}{1 + H_2 + H_1 H_2 H_3} R - G_1 E_1$

$Y = R \left(\frac{H_1 H_2 H_3}{1 + H_1 H_2 H_3 + H_2} - G_1 \left(\frac{1 + H_2}{1 + H_2 + H_1 H_2 H_3} \right) \right)$

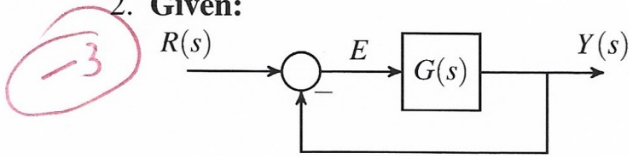
$\frac{Y}{R} = \frac{H_1 H_2 H_3 - G_1 (1 + H_2)}{1 + H_1 H_2 H_3 + H_2}$

= (

$$\underline{\underline{E = R - Y}}$$

T-1105

2. Given:



$$\begin{aligned} E &= -E G + R, \quad Y = E G \\ \Rightarrow E(1 + G) &= R \Rightarrow \frac{E}{R} = \frac{1}{1 + G} \\ \Rightarrow \frac{Y}{R} &= \frac{G}{1 + G} = \frac{Y}{E} \cdot \frac{E}{R} \\ \frac{Y}{E} &= G \end{aligned}$$

where $G(s)$ is defined by the time-domain equation: $\ddot{y}(t) - 4\dot{y}(t) - 2\dot{e}(t) = -4e(t) - 3y(t)$

Find:

- the transfer function, $\frac{Y(s)}{R(s)} = \frac{B(s)}{A(s)}$
- from part A determine the loop gain, L

$$A.) \mathcal{L} [\ddot{y}(t) - 4\dot{y}(t) + 3y(t)] = \mathcal{L} [2\dot{e}(t) - 4e(t)]$$

$$\Rightarrow Y(s)(s^2 - 4s + 3) = E(s)(2s - 4) \Rightarrow \frac{Y(s)}{E(s)} = \frac{2(s-2)}{(s^2 - 4s + 3)} = (R(s) - Y(s))(2s - 4)$$

$$\Rightarrow Y(s)(s^2 - 4s + 3 + 2s - 4) = R(s)(2s - 4)$$

$$\Rightarrow Y(s)(s^2 - 2s - 1) = R(s)(2s - 4)$$

$$\Rightarrow \frac{Y(s)}{R(s)} = \boxed{\frac{2s - 4}{s^2 - 2s - 1}} \quad \checkmark$$

$$B.) \frac{Y}{R} = \frac{B}{A} = \frac{G}{1 + GH} = \frac{1}{\frac{1}{G} + H} ; \quad \frac{2(s-2)}{s^2 - 2s - 1} \cdot \frac{1}{\frac{1}{2(s-2)}} = \frac{1}{\frac{s^2 - 2s}{2s - 4} + \frac{-1}{2s - 4}}$$

$$\text{Let } \frac{1}{G} = \frac{s^2 - 2s}{2s - 4} \Rightarrow G = \frac{2(s-2)}{s(s-2)} ; \quad \text{Let } H = \frac{-1}{2s - 4}$$

$$\Rightarrow L(s) = G(s)H(s) = \frac{-2(s-2)}{2s(s-2)^2} = \frac{-s+2}{s(s^2 - 4s + 4)} = \boxed{\frac{-s+2}{s^3 - 4s^2 + 4s}} \quad \checkmark$$

T-0151

T-0:51

3. Given:

$$\frac{Y(s)}{R(s)} = \frac{s^2 + 37s + 36}{(s+12)(s^2 + 8s + 7)}$$

When the input is a set point,

Find:

A. $y(\infty)$ with the FVT & verify stability

B. $y(0)$

A.) $r(t) = \begin{cases} 0 & : t \leq 0 \\ 1 & : t \geq 0 \end{cases} \Rightarrow R(s) = \frac{1}{s}$

$$Y(s) = \frac{Y(s)}{R(s)} \cdot R(s) = \frac{s^2 + 37s + 36}{s(s+12)(s^2 + 8s + 7)}$$

$$\lim_{t \rightarrow \infty} y(t) = \lim_{s \rightarrow 0^+} s Y(s) = \lim_{s \rightarrow 0^+} \left(\frac{s^2 + 37s + 36}{(s+12)(s^2 + 8s + 7)} \right) = \frac{(36)}{(12)(7)} = \boxed{\frac{3}{7}}$$

$$\frac{Y(s)}{R(s)} = \frac{(s+36)(s+1)}{(s+12)(s+7)(s+1)} \rightarrow p_i \in \{-12, -7, -1\}$$

Since $\text{Re}\{p_i\} < 0 \quad \forall p_i$ system is ~~stable~~ BIBO stable

T-0:47
T-0:37

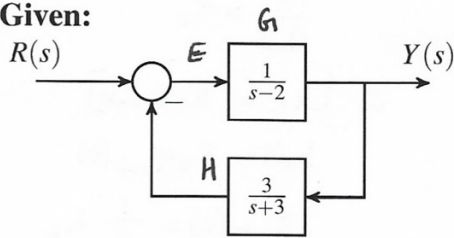
B.) ~~lim~~ $y(0) = \lim_{s \rightarrow \infty} s Y(s) = \lim_{s \rightarrow \infty} \left(\frac{s^2 + 37s + 36}{(s+12)(s^2 + 8s + 7)} \right) = \boxed{0}$

since degree of denominator (3) is greater than numerator

T-0:36

T-0147

4. Given:



Find:

- A. differential equation of the system for a generic input, $r(t)$
 B. the stability of the system for a bounded input

$$A.) \frac{Y}{R} = \frac{G_1}{1 + G_1 H} = \frac{\left(\frac{1}{s-2}\right)}{1 + \left(\frac{1}{s-2}\right)\left(\frac{3}{s+3}\right)} \cdot \frac{(s-2)(s+3)}{(s-2)(s+3)}$$

$$= \frac{(s+3)}{(s+3)(s-2) + 3} = \frac{(s+3)}{(s^2 + 3s - 2s - 6 + 3)} = \frac{(s+3)}{(s^2 + s - 3)}$$

$$\Rightarrow Y(s)(s^2 + s - 3) = R(s)(s+3)$$

$$\Rightarrow \mathcal{L}^{-1}[Y(s)(s^2 + s - 3)] = \mathcal{L}^{-1}[R(s)(s+3)]$$

$$\Rightarrow \boxed{\ddot{y}(t) + \dot{y}(t) - 3y(t) = \dot{r}(t) + 3r(t)}$$

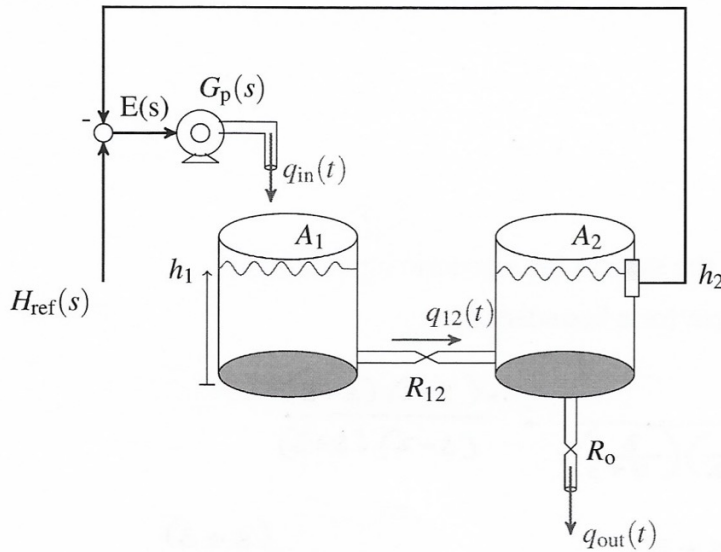
$$B.) \frac{Y}{R} = \frac{s+3}{s^2 + s - 3} \Rightarrow p_i : s = \frac{-1 \pm \sqrt{(1)^2 - 4(1)(-3)}}{2(1)} = \frac{-1 \pm \sqrt{13}}{2}$$

$$\Rightarrow p_i = \frac{-1 \pm \sqrt{13}}{2} \approx p_i \in \{-2.30, +1.30\}$$

Since not all of the roots have a negative real component
 this system is NOT BIBO STABLE

T-0139

5. Given the system described by the diagram and table below



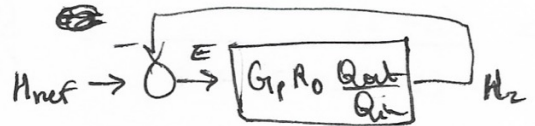
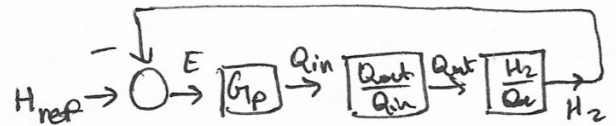
Parameter	Value	Identification
h_2		Height of fluid in tank 2
h_{ref}		Reference height
A_1	2	Area of tank 1
A_2	3	Area of tank 2
R_{12}	1	Restriction on $q_1(t)$
R_o	1	Restriction on $q_{out}(t)$
$G_p(s)$	$\frac{1}{s+1}$	Pump

$$\frac{H_2}{Q_{out}} = R_o$$

$$\mathcal{L}^{-1} [H_2 = R_o Q_{out}]$$

$$h_2(t) = R_o q_{out}(t)$$

$$q_{out}(t) = \frac{1}{R_o} h_2$$



$$\underline{H_2} =$$

Find: the transfer function $\frac{H_2(s)}{H_{ref}(s)}$

$$\frac{Q_{out}}{Q_{in}} = \left(A_1 A_2 R_{12} R_o s^2 + (A_1 R_{12} + A_2 R_o + A_1 R_o) s + 1 \right)^{-1} = \frac{1}{6s^2 + 7s + 1} = \frac{1}{(6s+1)(s+1)}$$

$$G_p = \frac{Q_{in}}{H_{ref}} \Rightarrow \frac{H_2}{H_{ref}} = \frac{Q_{in}}{H_{ref}} \cdot \frac{Q_{out}}{Q_{in}} \cdot \frac{H_2}{Q_{out}} = \frac{1}{(s+1)} \cdot \frac{1}{(6s+1)(s+1)} \cdot R_o$$

$$\frac{H_2(s)}{H_{ref}(s)} = \frac{1}{6s^3 + 7s^2 + s + 6s^2 + 7s + 1} = \boxed{\frac{1}{6s^3 + 13s^2 + 8s + 1}}$$

T-0:02

T-0:23

$$1.) A.) E_2(1+H_2) = E_1 H_1 \Rightarrow E_2(1+H_2) = (R - E_2 H_2 H_3) H_1$$

$$\Rightarrow E_2(1+H_2 + H_1 H_2 H_3) = R H_1 \Rightarrow E_2 = \frac{R H_1}{1 + H_1 H_2 H_3 + H_2}$$

$$\Rightarrow E_1 = R - \frac{H_2 H_3 R H_1}{1 + H_1 H_2 H_3 + H_2} = R \left(1 - \frac{H_1 H_2 H_3}{1 + H_2 + H_1 H_2 H_3} \right) = R \left(\frac{1 + H_2}{1 + H_2 + H_1 H_2 H_3} \right)$$

$$\Rightarrow Y = \frac{R H_1 H_2 H_3}{1 + H_1 H_2 H_3 + H_2} - \frac{R G_1 (1 + H_2)}{1 + H_1 H_2 H_3 + H_2}$$

$$\Rightarrow \frac{Y}{R} = \frac{H_1 H_2 H_3 - G_1 (1 + H_2)}{1 + H_1 H_2 H_3 + H_2} = \frac{\left(\frac{1}{s+1}\right)\left(\frac{3}{s-3}\right)\left(\frac{1}{s+2}\right) - \left(\frac{4}{s+4}\right)\left(1 + \frac{3}{s-3}\right)}{1 + \left(\frac{1}{s+1}\right)\left(\frac{3}{s-3}\right)\left(\frac{1}{s+2}\right) + \left(\frac{3}{s-3}\right)}$$

$$= \frac{3(s+4) - 4s(s+1)(s+2)}{(s+1)(s+2)(s-3)(s+4) + 3(s+4) + 3(s+1)(s+2)(s+4)}$$

$$= \frac{3s+12 - 4s^3 - 12s^2 - 8s}{s^4 + s^3 - 12s^2 + 3s^3 + 3s^2 - 36s + 2s^2 + 2s - 24 + 3s+12 + 3s^3 + 12s^2 + 9s^2 + 36s \dots + 6s + 24}$$

$$= \boxed{\frac{-4s^3 - 12s^2 - 5s + 12}{s^4 + 7s^3 + 14s^2 + 9s + 12}}$$

T-0:09

$$B.) \frac{1}{\frac{1}{G} + H}$$

$$G =$$

$$X \frac{-4s^3 - 12s^2 - 5s + 12}{s^4 + 7s^3 + 14s^2 + 9s}$$

$$H = \frac{12}{-4s^3 - 12s^2 - 5s + 12}$$

$$L = GH =$$

$$\frac{-48s^3 - 144s^2 - 60s + 144}{\dots}$$